

Отчёт о лабораторной работе

Лабораторная работа 8

Мошаров Денис Максимович

Содержание

Цель работы	3
Выполнение лабораторной работы	3
Самостоятельная работа.....	48
Выводы.....	65

Список иллюстраций

Новый проект	4
Топология.....	4
Настройка PC1.....	5
Настройка PC2.....	5
msk-dmmosharov-gw-01	6
msk-dmmosharov-gw-02	7
msk-dmmosharov-gw-03	8
msk-dmmosharov-gw-04	9
IPv6-PC1	9
IPv6-PC2	10
msk-dmmosharov-gw-01	11
Пmsk-dmmosharov-gw-02	12
msk-dmmosharov-gw-03	12
msk-dmmosharov-gw-04	15
msk-dmmosharov-gw-01	15
msk-dmmosharov-gw-02	16
msk-dmmosharov-gw-03	16
msk-dmmosharov-gw-04	17
Отсутствие блокировки	17
Проверка RIP	18
Проверка RIP	19
Проверка RIP	20
Проверка RIP	21
Пингуем PC2 с PC1	22
Метрики	22
Отключение интерфейса	22
Метрики	23
С PC1 пингуем PC2	23
Включим на маршрутизаторе	23

С PC1 пингуем PC2	24
Настраиваем RIPng для msk-dmmosharov-gw-01.....	25
Настраиваем RIPng для msk-dmmosharov-gw-02.....	26
Настраиваем RIPng для msk-dmmosharov-gw-03.....	26
Настраиваем RIPng для msk-dmmosharov-gw-04.....	27
С PC1 пропингуем PC2	27
Проверяем метрики.....	28
Отключаем интерфейс.....	28
Проверяем метрики.....	28
Настроим OSPFv2 для msk-dmmosharov-gw-01.....	30
Настроим OSPFv2 для msk-dmmosharov-gw-02.....	30
Настроим OSPFv2 для msk-dmmosharov-gw-03.....	30
Настроим OSPFv2 для msk-dmmosharov-gw-03.....	31
С PC1 пропингуем PC2	31
Проверка таблицы.....	32
С PC1 пропингуем PC2	32
Включаем интерфейс.....	33
Захваченный трафик.....	33
настроим OSPFv3 для сетей IPv6	34
настроим OSPFv3 для сетей IPv6	34
настроим OSPFv3 для сетей IPv6	35
настроим OSPFv3 для сетей IPv6	35
С PC1 пропингуем PC2	36
Проверка таблицы.....	36
С PC1 пропингуем PC2	37
Захваченный трафик.....	37
Проект	38
Присвоим адреса.....	38
Присвоим адреса.....	39
Перейдем в режим конфигурирования	39
Перейдем в режим конфигурирования	40
Перейдем в режим конфигурирования	40
Настроим адреса.....	41
Настроим адреса.....	41
Настроим адреса.....	42
Убедимся, что появились адреса.....	42
Настройка маршрутизации	43
Настройка маршрутизации	43
Настройка маршрутизации	44
Проверка	45
Туннель	46
Туннель	46
Настройка.....	47
Настройка.....	47
Проверка доступности.....	47
Проверка доступности.....	48

Трафик	48
Топология сети	49
Самостоятельная работа.....	49
Самостоятельная работа.....	50
Самостоятельная работа.....	51
Самостоятельная работа.....	52
Самостоятельная работа.....	53
Самостоятельная работа.....	54
Самостоятельная работа.....	55
Самостоятельная работа.....	56
Самостоятельная работа.....	57
Самостоятельная работа.....	58
Самостоятельная работа.....	59
Самостоятельная работа.....	60
Самостоятельная работа.....	61
Самостоятельная работа.....	61
Самостоятельная работа.....	61
Самостоятельная работа.....	62
Самостоятельная работа.....	63
Самостоятельная работа.....	64
Самостоятельная работа.....	65

Список таблиц

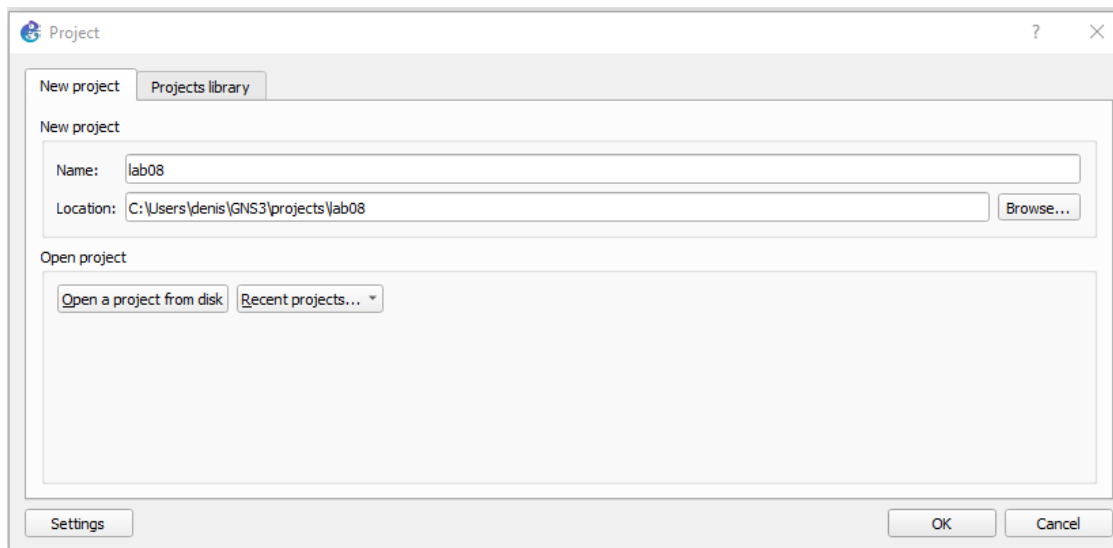
No table of figures entries found.

Цель работы

Изучить принципы маршрутизации в IPv4- и IPv6-сетях и принципы настройки сетевого оборудования.

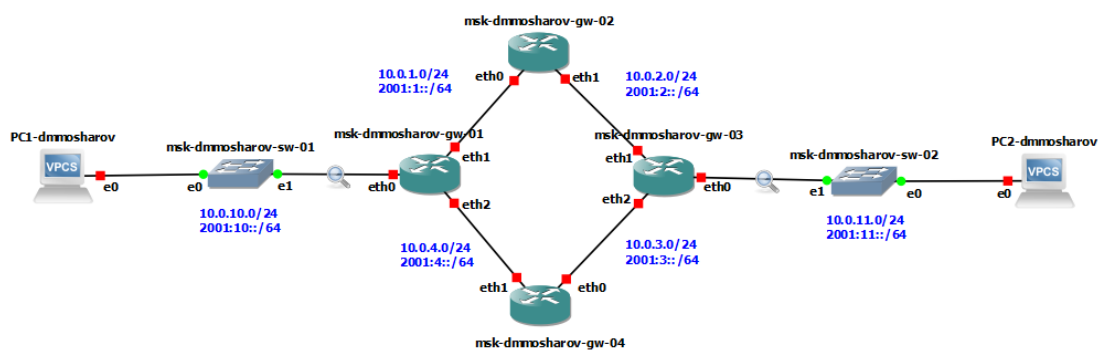
Выполнение лабораторной работы

Создаём новый проект в GNS3 с названием “lab8” (рис. [-@fig:001]).



Новый проект

Выстраиваем топологию сети, согласно данному заданию, размещаем коммутаторы, маршрутизаторы FRR и оконечные устройства (рис. [-@fig:002]).



Топология

Настраиваем IP-адрес и шлюз по умолчанию на PC1 10.0.10.10/24 10.0.10.1 (рис. [-@fig:003]).

```

VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.0.10.10/24
GATEWAY     : 10.0.10.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20064
RHOST:PORT  : 127.0.0.1:20065
MTU         : 1500

VPCS> █

```

Настройка PC1

Настраиваем IP-адрес и шлюз по умолчанию на PC2 ip 10.0.11.10/24 10.0.11.1 (рис. [-@fig:004]).

```

VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.0.11.10/24
GATEWAY     : 10.0.11.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20082
RHOST:PORT  : 127.0.0.1:20083
MTU         : 1500

VPCS> █

```

Настройка PC2

Настроим IPv4-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-01 (рис. [-@fig:005]).

```

frr# configure terminal
frr(config)# hostname msk-dmmosharov-gw-01
msk-dmmosharov-gw-01(config)# interface eth0
msk-dmmosharov-gw-01(config-if)# ip address 10.0.10.1/24
msk-dmmosharov-gw-01(config-if)# no shutdown
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# interface eth1
msk-dmmosharov-gw-01(config-if)# ip address 10.0.1.1/24
msk-dmmosharov-gw-01(config-if)# no shutdown
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# interface eth2
msk-dmmosharov-gw-01(config-if)# ip address 10.0.4.2/24
msk-dmmosharov-gw-01(config-if)# no shutdown
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# exit
msk-dmmosharov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-dmmosharov-gw-01# 

```

msk-dmmosharov-gw-01

Настроим IPv4-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-02 (рис. [-@fig:006]).

```
frr# configure terminal
frr(config)# hostname msk-dmmosharov-gw-02
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# ip address 10.0.1.2/24
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# interface eth1
msk-dmmosharov-gw-02(config-if)# ip address 10.0.2.1/24
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
 ip address 10.0.2.1/24
exit
!
end
msk-dmmosharov-gw-02#
```

msk-dmmosharov-gw-02

Настроим IPv4-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-03 (рис. [-@fig:007]).

```

frr# configure terminal
frr(config)# hostname msk-dmmosharov-gw-03
msk-dmmosharov-gw-03(config)# interface eth0
msk-dmmosharov-gw-03(config-if)# ip address 10.0.11.1/24
msk-dmmosharov-gw-03(config-if)# no shutdown
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# interface eth1
msk-dmmosharov-gw-03(config-if)# ip address 10.0.2.2/24
msk-dmmosharov-gw-03(config-if)# no shutdown
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# interface eth2
msk-dmmosharov-gw-03(config-if)# ip address 10.0.3.1/24
msk-dmmosharov-gw-03(config-if)# no shutdown
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# exit
msk-dmmosharov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-dmmosharov-gw-03# █

```

msk-dmmosharov-gw-03

Настроим IPv4-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-04 (рис. [-@fig:008]).

```

frr# configure terminal
frr(config)# hostname msk-dmmosharov-gw-04
msk-dmmosharov-gw-04(config)# interface eth0
msk-dmmosharov-gw-04(config-if)# ip address 10.0.3.2/24
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# interface eth1
msk-dmmosharov-gw-04(config-if)# ip address 10.0.4.1/24
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# write memory
% Unknown command: write memory
msk-dmmosharov-gw-04(config)# show running-config
% Unknown command: show running-config
msk-dmmosharov-gw-04(config)# 

```

msk-dmmosharov-gw-04

Присваиваем IPv6-адреса оконечным устройствам PC1 и PC2 в соответствии с данными в таблице (рис. [-@fig:009]).

```

VPCS> ip 2001:10::a/64
PC1 : 2001:10::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 20064
RHOST:PORT          : 127.0.0.1:20065
MTU                 : 1500

VPCS> 

```

IPv6-PC1

Продельываем тоже самое для PC2 (рис. [-@fig:010]).

```
VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 20082
RHOST:PORT           : 127.0.0.1:20083
MTU                 : 1500

VPCS> █
```

IPv6-PC2

Настраиваем IPv6-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-01 (рис. [-@fig:011]).

```
msk-dmmosharov-gw-01# configure terminal
msk-dmmosharov-gw-01(config)# ipv6 forwarding
msk-dmmosharov-gw-01(config)# interface eth0
msk-dmmosharov-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-dmmosharov-gw-01(config-if)# no ipv6 nd suppress-ra
msk-dmmosharov-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-dmmosharov-gw-01(config-if)# no shutdown
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# interface eth1
msk-dmmosharov-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-dmmosharov-gw-01(config-if)# no shutdown
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# interface eth2
msk-dmmosharov-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-dmmosharov-gw-01(config-if)# no shutdown
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# exit
msk-dmmosharov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
 ipv6 address 2001:10::1/64
 ipv6 nd prefix 2001:10::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.1.1/24
 ipv6 address 2001:1::1/64
exit
!
interface eth2
 ip address 10.0.4.2/24
 ipv6 address 2001:4::2/64
exit
!
lines 1-23
```

msk-dmmosharov-gw-01

Настраиваем IPv6-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-02 (рис. [-@fig:011.1]).

Пmsk-dmmosharov-gw-02

Пmsk-dmmosharov-gw-02

Настраиваем IPv6-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-03 (рис. [-@fig:012]).

```
msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# ipv6 forwarding
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# interface eth1
msk-dmmosharov-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ip address 10.0.2.1/24
 ipv6 address 2001:2::1/64
exit
!
end
msk-dmmosharov-gw-02#
```

msk-dmmosharov-gw-03

Настраиваем IPv6-адреса на интерфейсах маршрутизаторов: msk-dmmosharov-gw-04 (рис. [-@fig:013]).

```
msk-dmmosharov-gw-03# configure terminal
msk-dmmosharov-gw-03(config)# ipv6 forwarding
msk-dmmosharov-gw-03(config)# interface eth0
msk-dmmosharov-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-dmmosharov-gw-03(config-if)# no ipv6 nd suppress-ra
msk-dmmosharov-gw-03(config-if)# ipv6 nd prefix 2001:11:/64
% Unknown command: ipv6 nd prefix 2001:11:/64
msk-dmmosharov-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-dmmosharov-gw-03(config-if)# no shutdown
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# interface eth1
msk-dmmosharov-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-dmmosharov-gw-03(config-if)# no shutdown
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# interface eth2
msk-dmmosharov-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-dmmosharov-gw-03(config-if)# no shutdown
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# exit
msk-dmmosharov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
 ipv6 address 2001:11::1/64
 ipv6 nd prefix 2001:11::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.2.2/24
 ipv6 address 2001:2::2/64
exit
!
interface eth2
 ip address 10.0.3.1/24
 ipv6 address 2001:3::1/64
exit
!
lines 1-23
```

msk-dmmosharov-gw-04

Настроим RIP на маршрутизаторах в качестве протокола динамической маршрутизации на каждом из msk-dmmosharov-gw-0* (рис. [-@fig:014]).

```
msk-dmmosharov-gw-04# configure terminal
msk-dmmosharov-gw-04(config)# ipv6 forwarding
msk-dmmosharov-gw-04(config)# interface eth0
msk-dmmosharov-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# interface eth1
msk-dmmosharov-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
exit
!
end
msk-dmmosharov-gw-04#
```

msk-dmmosharov-gw-01

Также протокол RIP версии 2 на следующем маршрутизаторе (рис. [-@fig:015]).

```
msk-dmmosharov-gw-01# configure terminal
msk-dmmosharov-gw-01(config)# router rip
msk-dmmosharov-gw-01(config-router)# version 2
msk-dmmosharov-gw-01(config-router)# network eth0
msk-dmmosharov-gw-01(config-router)# network eth1
msk-dmmosharov-gw-01(config-router)# network eth2
msk-dmmosharov-gw-01(config-router)# exit
msk-dmmosharov-gw-01(config)# exit
msk-dmmosharov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-01#
```

msk-dmmosharov-gw-02

Также протокол RIP версии 2 на следующем маршрутизаторе (рис. [-@fig:016]).

```
msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# router rip
msk-dmmosharov-gw-02(config-router)# version 2
msk-dmmosharov-gw-02(config-router)# network eth0
msk-dmmosharov-gw-02(config-router)# network eth1
msk-dmmosharov-gw-02(config-router)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02#
```

msk-dmmosharov-gw-03

Также протокол RIP версии 2 на следующем маршрутизаторе (рис. [-@fig:017]).

```
msk-dmmosharov-gw-03# configure terminal
msk-dmmosharov-gw-03(config)# router rip
msk-dmmosharov-gw-03(config-router)# version 2
msk-dmmosharov-gw-03(config-router)# network eth0
msk-dmmosharov-gw-03(config-router)# network eth1
msk-dmmosharov-gw-03(config-router)# network eth2
msk-dmmosharov-gw-03(config-router)# exit
msk-dmmosharov-gw-03(config)# exit
msk-dmmosharov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-03#
```

msk-dmmosharov-gw-04

Выведя статус защиты SSH убеждаемся, что бан был снят (рис. [-@fig:018]).

```
msk-dmmosharov-gw-04# configure terminal
msk-dmmosharov-gw-04(config)# router rip
msk-dmmosharov-gw-04(config-router)# version 2
msk-dmmosharov-gw-04(config-router)# network eth0
msk-dmmosharov-gw-04(config-router)# network eth1
msk-dmmosharov-gw-04(config-router)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04#
```

Отсутствие блокировки

Убеждаемся, что маршрутизация по RIP настроена на каждом маршрутизаторе (рис. [-@fig:019]).

```

msk-dmmosharov-gw-01# ip route rip
% Unknown command: ip route rip
msk-dmmosharov-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/2] via 10.0.1.2, eth1, weight 1, 00:03:59
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:02:11
R>* 10.0.11.0/24 [120/3] via 10.0.1.2, eth1, weight 1, 00:03:02
msk-dmmosharov-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

        Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24          0.0.0.0            1 self                0
R(n) 10.0.2.0/24          10.0.1.2           2 10.0.1.2             0 02:49
R(n) 10.0.3.0/24          10.0.4.1           2 10.0.4.1             0 02:53
C(i) 10.0.4.0/24          0.0.0.0            1 self                0
C(i) 10.0.10.0/24         0.0.0.0            1 self                0
R(n) 10.0.11.0/24         10.0.1.2           3 10.0.1.2             0 02:49
msk-dmmosharov-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 6 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0            2     2
    eth1            2     2
    eth2            2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway          BadPackets  BadRoutes  Distance  Last Update
    10.0.1.2          0           0          120       00:00:15
    10.0.4.1          0           0          120       00:00:11
  Distance: (default is 120)
msk-dmmosharov-gw-01# 

```

Проверка RIP

Проверяем gw-02 (рис. [-@fig:020]).

```

msk-dmmosharov-gw-02# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.3.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:04:37
R>* 10.0.4.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:05:20
R>* 10.0.10.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:05:20
R>* 10.0.11.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:04:20
msk-dmmosharov-gw-02# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

        Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24          0.0.0.0            1 self                0
C(i) 10.0.2.0/24          0.0.0.0            1 self                0
R(n) 10.0.3.0/24          10.0.2.2           2 10.0.2.2            0 02:56
R(n) 10.0.4.0/24          10.0.1.1           2 10.0.1.1            0 02:51
R(n) 10.0.10.0/24         10.0.1.1           2 10.0.1.1            0 02:51
R(n) 10.0.11.0/24         10.0.2.2           2 10.0.2.2            0 02:56
msk-dmmosharov-gw-02# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 11 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0            2     2
    eth1            2     2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway          BadPackets  BadRoutes  Distance  Last Update
    10.0.1.1          0           0          120       00:00:15
    10.0.2.2          0           0          120       00:00:10
  Distance: (default is 120)
msk-dmmosharov-gw-02# █

```

Проверка RIP

Проверка gw-03 (рис. [-@fig:021]).

```

msk-dmmosharov-gw-03# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.2.1, eth1, weight 1, 00:06:18
R>* 10.0.4.0/24 [120/2] via 10.0.3.2, eth2, weight 1, 00:05:40
R>* 10.0.10.0/24 [120/3] via 10.0.2.1, eth1, weight 1, 00:06:18
msk-dmmosharov-gw-03# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

      Network          Next Hop          Metric From          Tag Time
R(n) 10.0.1.0/24       10.0.2.1           2 10.0.2.1           0 02:39
C(i) 10.0.2.0/24       0.0.0.0            1 self               0
C(i) 10.0.3.0/24       0.0.0.0            1 self               0
R(n) 10.0.4.0/24       10.0.3.2           2 10.0.3.2           0 02:47
R(n) 10.0.10.0/24      10.0.2.1           3 10.0.2.1           0 02:39
C(i) 10.0.11.0/24      0.0.0.0            1 self               0
msk-dmmosharov-gw-03# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 22 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface    Send  Recv  Key-chain
    eth0         2     2
    eth1         2     2
    eth2         2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets  BadRoutes  Distance  Last Update
    10.0.2.1      0           0          120      00:00:24
    10.0.3.2      0           0          120      00:00:16
  Distance: (default is 120)
msk-dmmosharov-gw-03#

```

Проверка RIP

Проверка gw-04 (рис. [-@fig:022]).

```

msk-dmmosharov-gw-04# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:07:46
R>* 10.0.2.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:07:47
R>* 10.0.10.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:07:46
R>* 10.0.11.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:07:47
msk-dmmosharov-gw-04# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

      Network          Next Hop          Metric From          Tag Time
R(n) 10.0.1.0/24       10.0.4.2           2 10.0.4.2           0 02:56
R(n) 10.0.2.0/24       10.0.3.1           2 10.0.3.1           0 02:41
C(i) 10.0.3.0/24       0.0.0.0            1 self              0
C(i) 10.0.4.0/24       0.0.0.0            1 self              0
R(n) 10.0.10.0/24      10.0.4.2           2 10.0.4.2           0 02:56
R(n) 10.0.11.0/24      10.0.3.1           2 10.0.3.1           0 02:41
msk-dmmosharov-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 19 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2     2
    eth1           2     2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway      BadPackets  BadRoutes  Distance  Last Update
    10.0.3.1      0           0          120      00:00:21
    10.0.4.2      0           0          120      00:00:07
  Distance: (default is 120)
msk-dmmosharov-gw-04#

```

Проверка RIP

Проверим пути прохождения пакетов, с PC1 пингуем PC2 (рис. [-@fig:023]).

```

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=2.511 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=3.162 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=2.714 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=2.832 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=2.971 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    1.120 ms  0.407 ms  0.437 ms
 2  10.0.1.2     1.814 ms  0.798 ms  1.145 ms
 3  10.0.2.2     2.307 ms  1.226 ms  1.371 ms
 4  10.0.11.10   1.991 ms  2.065 ms  1.338 ms

VPCS> █

```

Пингаем PC2 с PC1

Проверяем метрики протокола RIP на gw-01 (рис. [-@fig:024]).

```

msk-dmmosharov-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network      Next Hop      Metric From      Tag Time
C(i) 10.0.1.0/24   0.0.0.0       1 self           0
R(n) 10.0.2.0/24   10.0.1.2      2 10.0.1.2        0 02:45
R(n) 10.0.3.0/24   10.0.4.1      2 10.0.4.1        0 02:33
C(i) 10.0.4.0/24   0.0.0.0       1 self           0
C(i) 10.0.10.0/24  0.0.0.0       1 self           0
R(n) 10.0.11.0/24  10.0.1.2      3 10.0.1.2        0 02:45
msk-dmmosharov-gw-01# █

```

Метрики

Отключаем на маршрутизаторе gw-02 интерфейс (рис. [-@fig:025]).

```

msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# shutdown
msk-dmmosharov-gw-02(config-if)# █

```

Отключение интерфейса

Проверяем метрики протокола RIP на gw-01 (рис. [-@fig:026]).

```
msk-dmmosharov-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface

Network        Next Hop        Metric From      Tag Time
C(i) 10.0.1.0/24 0.0.0.0         1 self          0
R(n) 10.0.2.0/24 10.0.1.2        2 10.0.1.2      0 02:32
R(n) 10.0.3.0/24 10.0.4.1        2 10.0.4.1      0 02:53
C(i) 10.0.4.0/24 0.0.0.0         1 self          0
C(i) 10.0.10.0/24 0.0.0.0         1 self          0
R(n) 10.0.11.0/24 10.0.1.2        3 10.0.1.2      0 02:32
msk-dmmosharov-gw-01#
```

Метрики

С PC1 пингуем PC2 и определяем (рис. [-@fig:027]).

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=4.184 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=3.490 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=2.705 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=2.943 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=2.632 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1   0.787 ms  0.442 ms  0.351 ms
 2  10.0.4.1    1.543 ms  0.908 ms  1.166 ms
 3  10.0.3.1    1.818 ms  1.397 ms  2.032 ms
 4  10.0.11.10  1.851 ms  1.564 ms  1.864 ms

VPCS>
```

С PC1 пингуем PC2

Включим на маршрутизаторе msk-user-dmmosharov-02 интерфейс (рис. [-@fig:028]).

```
% Unknown command: infterface eth0
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)#
```

Включим на маршрутизаторе

С PC1 пропингуем PC2 и определим путь следования пакетов (рис. [-@fig:029]).

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=2.773 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=2.878 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=3.016 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=3.055 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=2.852 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    0.719 ms  0.609 ms  0.394 ms
 2  10.0.4.1     1.258 ms  0.790 ms  0.752 ms
 3  10.0.3.1     1.718 ms  1.279 ms  1.505 ms
 4  10.0.11.10   1.605 ms  1.984 ms  1.562 ms

VPCS> █
```

С PC1 пингуем PC2

Посмотрим захваченный на соединениях трафик (рис. [-@fig:030]).

No.	Time	Source	Destination	Protocol	Length	Info
376	9141.8925...	0c:95:11:9d:00...	Private_66:68:...	ARP	60	10.0.10.1 is at 0c:95:11:9d:00:00
377	9141.8932...	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x9135, seq=1/256, ttl=64 (reply in 378)
378	9141.8970...	10.0.10.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x9135, seq=1/256, ttl=61 (request in 377)
379	9142.8981...	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x9235, seq=2/512, ttl=64 (reply in 380)
380	9142.9000...	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x9235, seq=2/512, ttl=61 (request in 379)
381	9143.3544...	10.0.10.1	224.0.0.9	RIPv2	146	Response
382	9143.9020...	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x9335, seq=3/768, ttl=64 (reply in 383)
383	9143.9045...	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x9335, seq=3/768, ttl=61 (request in 382)
384	9144.9060...	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x9435, seq=4/1024, ttl=64 (reply in 385)
385	9144.9088...	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x9435, seq=4/1024, ttl=61 (request in 384)
386	9145.9096...	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x9535, seq=5/1280, ttl=64 (reply in 387)
387	9145.9118...	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x9535, seq=5/1280, ttl=61 (request in 386)
388	9146.9509...	0c:95:11:9d:00...	Private_66:68:...	ARP	60	Who has 10.0.10.10? Tell 10.0.10.1
389	9146.9510...	Private_66:68:...	0c:95:11:9d:00...	ARP	60	10.0.10.10 is at 00:50:79:66:68:00
390	9148.6115...	10.0.10.10	10.0.11.10	TCP	74	35008 → 35009 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766077848 TSecr=0 WS=2
391	9148.6122...	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
392	9148.6128...	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 35008 → 35009 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766077848 TSecr=0 WS=2
393	9148.6133...	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
394	9148.6130...	10.0.11.10	10.0.10.10	TCP	74	[TCP Retransmission] 35008 → 35009 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766077848 TSecr=0 WS=2
395	9148.6141...	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
396	9148.6154...	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 35008 → 35009 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766077848 TSecr=0 WS=2
397	9148.6162...	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
398	9148.6169...	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 35008 → 35009 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766077848 TSecr=0 WS=2
399	9148.6177...	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff) Address Resolution Protocol (request/gratuitous ARP)						
					0000	ff ff ff ff ff ff 00 50 79 66 68 00 08 06 00 01
					0010	08 00 06 04 00 01 00 50 79 66 68 00 0a 00 0a 0a
					0020	ff ff ff ff ff ff 0a 00 0a 0a 00 00 00 00 00 00
					0030	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff) Address Resolution Protocol (request/gratuitous ARP)						
					0000	ff ff ff ff ff ff 00 50 79 66 68 00 08 06 00 01
					0010	08 00 06 04 00 01 00 50 79 66 68 00 0a 00 0a 0a
					0020	ff ff ff ff ff ff 0a 00 0a 0a 00 00 00 00 00 00
					0030	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

На маршрутизаторах настроим RIPng для сетей IPv6: msk-dmmosharov-gw-01 (рис. [-@fig:032]).

```
msk-dmmosharov-gw-01# configure terminal
msk-dmmosharov-gw-01(config)# router ripng
msk-dmmosharov-gw-01(config-router)# network eth0
msk-dmmosharov-gw-01(config-router)# network eth1
msk-dmmosharov-gw-01(config-router)# network eth2
msk-dmmosharov-gw-01(config-router)# exit
msk-dmmosharov-gw-01(config)# exit
msk-dmmosharov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-01#
```

Настраиваем RIPng для msk-dmmosharov-gw-01

На маршрутизаторах настроим RIPng для сетей IPv6: msk-dmmosharov-gw-02 (рис. [-@fig:033]).

```
msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# infterface eth0
% Unknown command: infterface eth0
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# configure terminal
% Unknown command: configure terminal
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# router ripng
msk-dmmosharov-gw-02(config-router)# network eth0
msk-dmmosharov-gw-02(config-router)# network eth1
msk-dmmosharov-gw-02(config-router)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02#
```

Настраиваем RIPng для msk-dmmosharov-gw-02

На маршрутизаторах настроим RIPng для сетей IPv6: msk-dmmosharov-gw-03 (рис. [-@fig:034]).

```
msk-dmmosharov-gw-03# configure terminal
msk-dmmosharov-gw-03(config)# router ripng
msk-dmmosharov-gw-03(config-router)# network eth0
msk-dmmosharov-gw-03(config-router)# network eth1
msk-dmmosharov-gw-03(config-router)# network eth2
msk-dmmosharov-gw-03(config-router)# exit
msk-dmmosharov-gw-03(config)# exit
msk-dmmosharov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-03#
```

Настраиваем RIPng для msk-dmmosharov-gw-03

На маршрутизаторах настроим RIPng для сетей IPv6: msk-dmmosharov-gw-04 (рис. [-@fig:035]).

```

msk-dmmosharov-gw-04# configure terminal
msk-dmmosharov-gw-04(config)# router ripng
msk-dmmosharov-gw-04(config-router)# network eth0
msk-dmmosharov-gw-04(config-router)# network eth1
msk-dmmosharov-gw-04(config-router)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04# █

```

Настраиваем RIPng для msk-dmmosharov-gw-04

Проверим пути прохождения пакетов. С PC1 пропингуем PC2 и определим путь следования пакетов (рис. [-@fig:036]).

```

VPCS> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=3.195 ms
2001:11::a icmp6_seq=2 ttl=58 time=3.262 ms
2001:11::a icmp6_seq=3 ttl=58 time=2.566 ms
2001:11::a icmp6_seq=4 ttl=58 time=2.636 ms
2001:11::a icmp6_seq=5 ttl=58 time=3.054 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1  0.816 ms  0.538 ms  0.410 ms
 2 2001:1::2   1.390 ms  0.893 ms  0.761 ms
 3 2001:2::2   1.725 ms  1.419 ms  1.464 ms
 4 2001:11::a  1.954 ms  1.825 ms  1.390 ms

VPCS> █

```

С PC1 пропингуем PC2

Проверим метрики протокола RIPng (рис. [-@fig:037]).

```
msk-dmmosharov-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface, (a/S) - aggregated/Suppressed
```

	Network	Next Hop	Via	Metric	Tag	Time
C(i)	2001:1::/64	::	self	1	0	
R(n)	2001:2::/64	fe80::e18:3bff:fe59:0	eth1	2	0	02:28
R(n)	2001:3::/64	fe80::ed5:8cff:fe08:1	eth2	2	0	02:46
C(i)	2001:4::/64	::	self	1	0	
C(i)	2001:10::/64	::	self	1	0	
R(n)	2001:11::/64	fe80::e18:3bff:fe59:0	eth1	3	0	02:46

```
msk-dmmosharov-gw-01#
```

Проверяем метрики

Отключим на маршрутизаторе msk-dmmosharov-gw-02 интерфейс (рис. [-@fig:038]).

```
msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# shutdown
msk-dmmosharov-gw-02(config-if)#
```

Отключаем интерфейс

Проверим метрики протокола RIPng (рис. [-@fig:039]).

```
msk-dmmosharov-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface, (a/S) - aggregated/Suppressed
```

	Network	Next Hop	Via	Metric	Tag	Time
C(i)	2001:1::/64	::	self	1	0	
R(n)	2001:2::/64	fe80::e18:3bff:fe59:0	eth1	2	0	02:15
R(n)	2001:3::/64	fe80::ed5:8cff:fe08:1	eth2	2	0	02:40
C(i)	2001:4::/64	::	self	1	0	
C(i)	2001:10::/64	::	self	1	0	
R(n)	2001:11::/64	fe80::e18:3bff:fe59:0	eth1	3	0	02:40

```
msk-dmmosharov-gw-01#
```

Проверяем метрики

Посмотрим захваченный на соединениях трафик (рис. [-@fig:040]).

Захват из Standard input [msk-dmmosharov-sw-01 Ethernet1 to msk-dmmosharov-gw-01 eth0]																							
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка																							
Примените фильтр отображения ... <Ctrl-/>																							
No.	Time	Source	Destination	Protocol	Length	Info																	
797	15082.979.	10.0.10.1	224.0.0.9	RIPv2	146	Response																	
798	15083.014.	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xc64c, seq=3/768, ttl=64 (reply in 799)																	
799	15083.016.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc64c, seq=3/768, ttl=61 (request in 798)																	
800	15084.018.	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xc74c, seq=4/1024, ttl=64 (reply in 801)																	
801	15084.020.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc74c, seq=4/1024, ttl=61 (request in 800)																	
802	15085.022.	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xc84c, seq=5/1280, ttl=64 (reply in 803)																	
803	15085.024.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc84c, seq=5/1280, ttl=61 (request in 802)																	
804	15086.051.	0c:95:11:9d:00:...	Private 66:68:...	ARP	60	Who has 10.0.10.10? Tell 10.0.10.1																	
805	15086.051.	Private 66:68:...	0c:95:11:9d:00:...	ARP	60	10.0.10.10 is at 00:50:79:66:68:00																	
806	15094.095.	fe80::e95:11ff:...	ff02::9	RIPng	186	Command Response, Version 1																	
807	15098.684.	10.0.10.10	10.0.11.10	TCP	74	9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
808	15098.684.	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
809	15098.685.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
810	15098.685.	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
811	15098.687.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
812	15098.687.	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
813	15098.689.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
814	15098.690.	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
815	15098.691.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
816	15098.692.	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
817	15098.694.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
818	15098.694.	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
819	15098.695.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
820	15098.696.	10.0.3.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0						0000	ff	ff	ff	ff	ff	ff	00	50	79	66	68	00	08	06	00	01	
Ethernet II, Src: Private 66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)						0010	08	00	06	04	00	01	00	50	79	66	68	00	0a	00	0a	0a	0a
Address Resolution Protocol (request/gratuitous ARP)						0020	ff	ff	ff	ff	ff	ff	0a	00	0a	0a	00	00	00	00	00	00	00
						0030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
668	15000.290.	0c:35:d5:eb:00:...	Private 66:68:...	ARP	60	10.0.11.1 is at 0c:35:d5:eb:00:00																	
669	15000.292.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc44c, seq=1/256, ttl=64 (request in 666)																	
670	15001.296.	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xc54c, seq=2/512, ttl=61 (reply in 671)																	
671	15001.296.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc54c, seq=2/512, ttl=64 (request in 670)																	
672	15002.299.	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xc64c, seq=3/768, ttl=61 (reply in 673)																	
673	15002.299.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc64c, seq=3/768, ttl=64 (request in 672)																	
674	15003.303.	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xc74c, seq=4/1024, ttl=61 (reply in 675)																	
675	15003.303.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc74c, seq=4/1024, ttl=64 (request in 674)																	
676	15003.500.	fe80::e35:d5ff:...	ff02::9	RIPng	186	Command Response, Version 1																	
677	15004.307.	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xc84c, seq=5/1280, ttl=61 (reply in 678)																	
678	15004.307.	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xc84c, seq=5/1280, ttl=64 (request in 677)																	
679	15005.314.	0c:35:d5:eb:00:...	Private 66:68:...	ARP	60	Who has 10.0.11.10? Tell 10.0.11.1																	
680	15005.314.	Private 66:68:...	0c:35:d5:eb:00:...	ARP	60	10.0.11.10 is at 00:50:79:66:68:01																	
681	15017.987.	10.0.10.10	10.0.11.10	TCP	74	9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
682	15017.987.	10.0.11.10	10.0.10.10	TCP	54	9345 → 9344 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0																	
683	15017.991.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
684	15017.991.	10.0.11.10	10.0.10.10	TCP	54	[TCP Port numbers reused] 9345 → 9344 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0																	
685	15017.993.	10.0.10.10	10.0.11.10	TCP	74	9344 → 9345 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083798 TSecr=0 WS=2																	
686	15017.993.	10.0.11.10	10.0.10.10	TCP	54	[TCP Port numbers reused] 9345 → 9344 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0																	
687	15027.842.	10.0.11.1	224.0.0.9	RIPv2	146	Response																	
688	15031.499.	fe80::e35:d5ff:...	ff02::9	RIPng	186	Command Response, Version 1																	
689	15048.501.	fe80::e35:d5ff:...	ff02::9	RIPng	186	Command Response, Version 1																	
690	15057.841.	10.0.11.1	224.0.0.9	RIPv2	146	Response																	
Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0						0000	ff	ff	ff	ff	ff	ff	00	50	79	66	68	01	08	06	00	01	
Ethernet II, Src: Private 66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)						0010	08	00	06	04	00	01	00	50	79	66	68	01	0a	00	0b	0a	
Address Resolution Protocol (request/gratuitous ARP)						0020	ff	ff	ff	ff	ff	ff	0a	00	0a	0a	00	00	00	00	00	00	00
						0030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
851	15210.556.	0c:95:11:9d:00:...	Private 66:68:...	ARP	60	Who has 10.0.10.10? Tell 10.0.10.1																	
852	15210.556.	Private 66:68:...	0c:95:11:9d:00:...	ARP	60	10.0.10.10 is at 00:50:79:66:68:00																	
853	15212.915.	10.0.10.10	10.0.11.10	TCP	74	12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
854	15212.916.	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
855	15212.917.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
856	15212.917.	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
857	15212.918.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
858	15212.919.	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
859	15212.919.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
860	15212.920.	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
861	15212.920.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
862	15212.922.	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
863	15212.922.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
864	15212.922.	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
865	15212.923.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
866	15212.925.	10.0.3.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
867	15212.926.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
868	15212.927.	10.0.3.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
869	15212.929.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
870	15212.930.	10.0.3.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)																	
871	15212.931.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
872	15212.933.	10.0.11.10	10.0.10.10	TCP	60	12593 → 12592 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0																	
873	15212.934.	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 12592 → 12593 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766083912 TSecr=0 WS=2																	
874	15212.936.	10.0.11.10	10.0.10.10	TCP	60	[TCP Port numbers reused] 12593 → 12592 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0																	

На маршрутизаторах настроим OSPFv2 для сетей IPv4: msk-dmmosharov-gw-01 (рис. [-@fig:043]).

```

msk-dmmosharov-gw-01# configure terminal
msk-dmmosharov-gw-01(config)# router ospf
msk-dmmosharov-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-dmmosharov-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-dmmosharov-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-dmmosharov-gw-01(config-router)# exit
msk-dmmosharov-gw-01(config)# exit
msk-dmmosharov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-01# █

```

Настроим OSPFv2 для msk-dmmosharov-gw-01

На маршрутизаторах настроим OSPFv2 для сетей IPv4: msk-dmmosharov-gw-02 (рис. [-@fig:044]).

```

rmsk-dmmosharov-gw-02# configure terminal
rmsk-dmmosharov-gw-02(config)# router ospf
rmsk-dmmosharov-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
rmsk-dmmosharov-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
rmsk-dmmosharov-gw-02(config-router)# exit
rmsk-dmmosharov-gw-02(config)# exit
rmsk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
rmsk-dmmosharov-gw-02# █

```

Настроим OSPFv2 для msk-dmmosharov-gw-02

На маршрутизаторах настроим OSPFv2 для сетей IPv4: msk-dmmosharov-gw-03 (рис. [-@fig:045]).

```

msk-dmmosharov-gw-03# configure terminal
msk-dmmosharov-gw-03(config)# router ospf
msk-dmmosharov-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-dmmosharov-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-dmmosharov-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-dmmosharov-gw-03(config-router)# exit
msk-dmmosharov-gw-03(config)# exit
msk-dmmosharov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-03# █

```

Настроим OSPFv2 для msk-dmmosharov-gw-03

На маршрутизаторах настроим OSPFv2 для сетей IPv4: msk-dmmosharov-gw-04 (рис. [-@fig:046]).

```

msk-dmmosharov-gw-04# configure terminal
msk-dmmosharov-gw-04(config)# router ospf
msk-dmmosharov-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-dmmosharov-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-dmmosharov-gw-04(config-router)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04# █

```

Настроим OSPFv2 для msk-dmmosharov-gw-03

С PC1 пропингуем PC2 и определим путь следования пакетов (рис. [-@fig:047]).

```

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=4.204 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=2.110 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=2.847 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=2.537 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=2.395 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1   0.594 ms  0.390 ms  0.340 ms
 2  10.0.1.2    1.130 ms  0.878 ms  0.808 ms
 3  10.0.2.2    1.652 ms  1.243 ms  1.245 ms
 4  10.0.11.10  1.666 ms  1.737 ms  1.708 ms

VPCS> █

```

С PC1 пропингуем PC2

Проверим таблицу маршрутизации протокола (рис. [-@fig:048]).

```
msk-dmmosharov-gw-01# show ip ospf neighbor

Neighbor ID      Pri State                Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1         1 Full/Backup          3m12s             38.146s 10.0.1.2          et
h1:10.0.1.1      0      0      0
10.0.4.1         1 Full/Backup          1m02s             35.545s 10.0.4.1          et
h2:10.0.4.2      0      0      0

msk-dmmosharov-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24          [100] area: 0.0.0.0
    directly attached to eth1
N   10.0.2.0/24          [200] area: 0.0.0.0
    via 10.0.1.2, eth1
N   10.0.3.0/24          [200] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.4.0/24          [100] area: 0.0.0.0
    directly attached to eth2
N   10.0.10.0/24         [100] area: 0.0.0.0
    directly attached to eth0
N   10.0.11.0/24         [300] area: 0.0.0.0
    via 10.0.1.2, eth1
    via 10.0.4.1, eth2

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-dmmosharov-gw-01#
```

Проверка таблицы

С PC1 пропингуем PC2 и определим путь следования пакетов (рис. [-@fig:050]).

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=4.009 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=2.299 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=2.585 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=3.174 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=2.556 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    0.731 ms  0.513 ms  0.457 ms
 2  10.0.4.1     0.862 ms  1.284 ms  0.827 ms
 3  10.0.3.1     1.400 ms  1.086 ms  1.395 ms
 4  10.0.11.10   1.730 ms  1.371 ms  1.719 ms

VPCS>
```

С PC1 пропингуем PC2

Включим на маршрутизаторе msk-dmmosharov-gw-02 интерфейс (рис. [-@fig:051]).

```

msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# 

```

Включаем интерфейс

Посмотрим захваченный на соединениях трафик (рис. [-@fig:052]).

The image shows a Wireshark packet capture window. The top pane displays a list of captured packets. The middle pane shows the details of the selected packet (No. 1079), which is a TCP SYN packet from 10.0.10.10 to 10.0.11.10. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1063	15785.752	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1064	15785.753	10.0.10.10	10.0.11.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
1065	15785.753	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1066	15785.754	10.0.10.10	10.0.11.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
1067	15785.754	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1068	15785.755	10.0.10.10	10.0.11.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
1069	15785.756	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1070	15785.757	10.0.10.10	10.0.11.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
1071	15785.758	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1072	15785.760	10.0.10.10	10.0.11.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
1073	15785.761	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1074	15785.762	10.0.10.10	10.0.11.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
1075	15785.762	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1076	15785.764	10.0.10.10	10.0.11.10	TCP	60	50242 → 50241 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1077	15785.764	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1078	15785.766	10.0.10.10	10.0.11.10	TCP	60	[TCP Port numbers reused] 50242 → 50241 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1079	15785.767	10.0.10.10	10.0.11.10	TCP	74	50241 → 50242 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766084485 TSecr=0 WS=2
1080	15785.768	10.0.10.10	10.0.11.10	TCP	60	[TCP Port numbers reused] 50242 → 50241 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1081	15786.921	10.0.10.1	224.0.0.9	RIPv2	66	Response
1082	15787.218	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1083	15791.072	10.0.10.1	224.0.0.9	RIPv2	146	Response
1084	15791.185	fe80::e95:11ff...	ff02::9	RIPng	186	Command Response, Version 1
1085	15797.218	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet

Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
 Ethernet II, Src: Private, 66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 Address Resolution Protocol (request/gratuitous ARP)

0000 ff ff ff ff ff ff 00 50 79 66 68 00 00 00 00 01
 0010 00 00 00 04 00 01 00 50 79 66 68 00 0a 00 0a 0a
 0020 ff ff ff ff ff 0a 00 0a 0a 00 00 00 00 00 00
 0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Захваченный трафик

На маршрутизаторах настроим OSPFv3 для сетей IPv6: msk-dmmosharov-gw-01 (рис. [-@fig:053]).

```

msk-dmmosharov-gw-01# configure terminal
msk-dmmosharov-gw-01(config)# router ospf6
msk-dmmosharov-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-dmmosharov-gw-01(config-ospf6)# exit
msk-dmmosharov-gw-01(config)# interface eth0
msk-dmmosharov-gw-01(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# interface eth1
msk-dmmosharov-gw-01(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# interface eth2
msk-dmmosharov-gw-01(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-01(config-if)# exit
msk-dmmosharov-gw-01(config)# exit
msk-dmmosharov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-01# █

```

настроим OSPFv3 для сетей IPv6

На маршрутизаторах настроим OSPFv3 для сетей IPv6: msk-dmmosharov-gw-02 (рис. [-@fig:054]).

```

msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# router ospf6
msk-dmmosharov-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-dmmosharov-gw-02(config-ospf6)# exit
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# interface eth1
msk-dmmosharov-gw-02(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02# █

```

настроим OSPFv3 для сетей IPv6

На маршрутизаторах настроим OSPFv3 для сетей IPv6: msk-dmmosharov-gw-03 (рис. [-@fig:055]).

```

msk-dmmosharov-gw-03# configure terminal
msk-dmmosharov-gw-03(config)# router ospf6
msk-dmmosharov-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-dmmosharov-gw-03(config-ospf6)# exit
msk-dmmosharov-gw-03(config)# interface eth0
msk-dmmosharov-gw-03(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# interface eth1
msk-dmmosharov-gw-03(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# interface eth2
msk-dmmosharov-gw-03(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-03(config-if)# exit
msk-dmmosharov-gw-03(config)# exit
msk-dmmosharov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-03#

```

настроим OSPFv3 для сетей IPv6

На маршрутизаторах настроим OSPFv3 для сетей IPv6: msk-dmmosharov-gw-04 (рис. [-@fig:056]).

```

msk-dmmosharov-gw-04# configure terminal
msk-dmmosharov-gw-04(config)# router ospf6
msk-dmmosharov-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-dmmosharov-gw-04(config-ospf6)# exit
msk-dmmosharov-gw-04(config)# interface eth0
msk-dmmosharov-gw-04(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# interface eth1
msk-dmmosharov-gw-04(config-if)# ipv6 ospf6 area 0
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04#

```

настроим OSPFv3 для сетей IPv6

С PC1 пропингуем PC2 и определим путь следования пакетов (рис. [-@fig:057]).

```

VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=2.494 ms
2001:11::a icmp6_seq=2 ttl=58 time=3.139 ms
tr2001:11::a icmp6_seq=3 ttl=58 time=3.138 ms
01:11::a icmp6_seq=4 ttl=58 time=2.142 ms
2001:11::a icmp6_seq=5 ttl=58 time=2.335 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    1.067 ms  0.273 ms  0.403 ms
 2 2001:1::2    1.562 ms  0.800 ms  1.355 ms
 3 2001:2::2    1.785 ms  1.671 ms  1.345 ms
 4 2001:11::a   2.728 ms  1.627 ms  1.413 ms

VPCS> 

```

С PC1 пропингуем PC2

Проверим таблицу маршрутизации протокола (рис. [-@fig:058]).

```

msk-dmmosharov-gw-01# show ipv6 ospf6 neighbor
Neighbor ID      Pri   DeadTime   State/IfState      Duration I/F[State]
2.2.2.2          1     00:00:33   Full/BDR           00:04:05 eth1[DR]
4.4.4.4          1     00:00:35   Full/BDR           00:00:51 eth2[DR]
msk-dmmosharov-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64      ::                eth1 00:00:57
*N IA 2001:2::/64      fe80::e18:3bff:fe59:0 eth1 00:00:57
*N IA 2001:3::/64      fe80::ed5:8cff:fe08:1 eth2 00:00:57
*N IA 2001:4::/64      ::                eth2 00:00:57
*N IA 2001:10::/64     ::                eth0 00:00:57
*N IA 2001:11::/64     fe80::e18:3bff:fe59:0 eth1 00:00:57
                       fe80::ed5:8cff:fe08:1 eth2
msk-dmmosharov-gw-01# 

```

Проверка таблицы

С PC1 пропингуем PC2 и определим путь следования пакетов (рис. [-@fig:060]).

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=2.545 ms
^[[A^[[A2001:11::a icmp6_seq=2 ttl=58 time=2.736 ms
2001:11::a icmp6_seq=3 ttl=58 time=2.589 ms
2001:11::a icmp6_seq=4 ttl=58 time=2.749 ms
2001:11::a icmp6_seq=5 ttl=58 time=2.534 ms

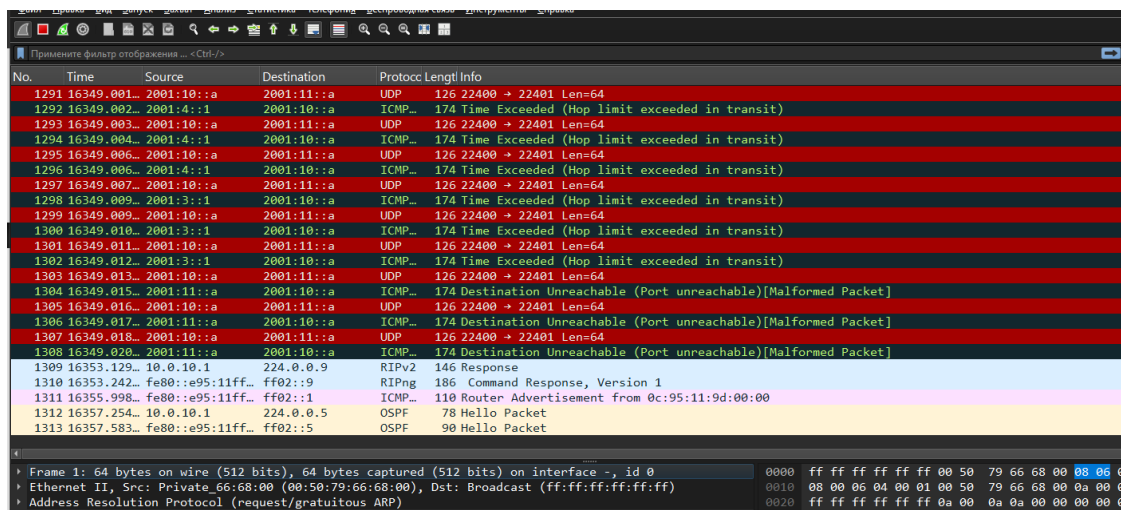
VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1 1.345 ms 0.404 ms 0.391 ms
 2 2001:4::1 1.186 ms 1.132 ms 0.928 ms
 3 2001:3::1 2.108 ms 1.194 ms 1.124 ms
 4 2001:11::a 1.846 ms 1.522 ms 1.577 ms

VPCS> █
```

С PC1 пропингуем PC2

Посмотрим захваченный трафик (рис. [-@fig:061]).

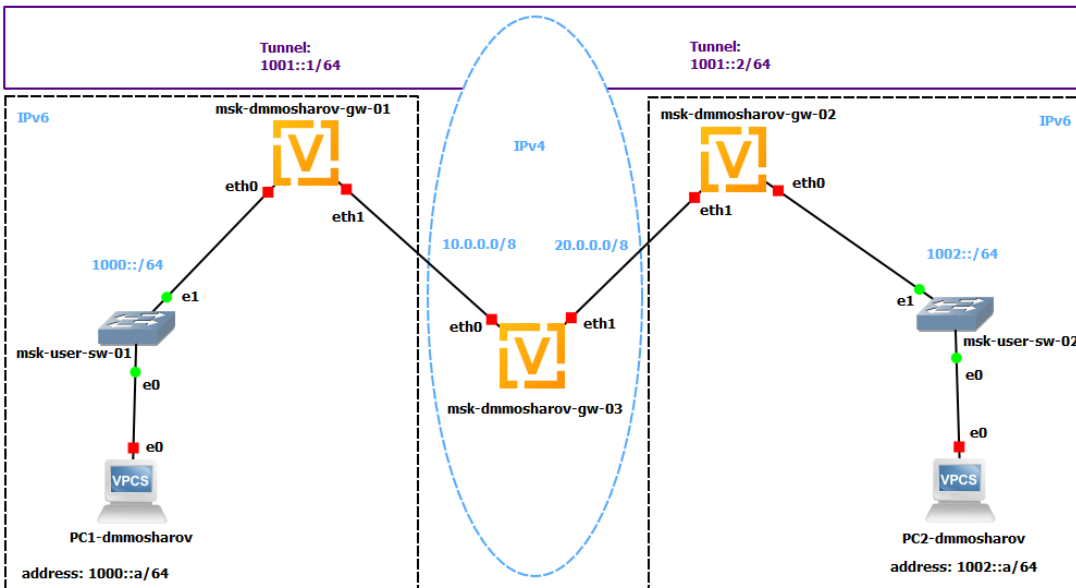


No.	Time	Source	Destination	Protocol	Length	Info
1291	16349.001	2001:10::1	2001:11::a	UDP	126	22400 → 22401 Len=64
1292	16349.002	2001:4::1	2001:10::a	ICMP...	174	Time Exceeded (Hop limit exceeded in transit)
1293	16349.003	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1294	16349.004	2001:4::1	2001:10::a	ICMP...	174	Time Exceeded (Hop limit exceeded in transit)
1295	16349.006	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1296	16349.006	2001:4::1	2001:10::a	ICMP...	174	Time Exceeded (Hop limit exceeded in transit)
1297	16349.007	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1298	16349.009	2001:3::1	2001:10::a	ICMP...	174	Time Exceeded (Hop limit exceeded in transit)
1299	16349.009	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1300	16349.010	2001:3::1	2001:10::a	ICMP...	174	Time Exceeded (Hop limit exceeded in transit)
1301	16349.011	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1302	16349.012	2001:3::1	2001:10::a	ICMP...	174	Time Exceeded (Hop limit exceeded in transit)
1303	16349.013	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1304	16349.015	2001:11::a	2001:10::a	ICMP...	174	Destination Unreachable (Port unreachable)[Malformed Packet]
1305	16349.016	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1306	16349.017	2001:11::a	2001:10::a	ICMP...	174	Destination Unreachable (Port unreachable)[Malformed Packet]
1307	16349.018	2001:10::a	2001:11::a	UDP	126	22400 → 22401 Len=64
1308	16349.020	2001:11::a	2001:10::a	ICMP...	174	Destination Unreachable (Port unreachable)[Malformed Packet]
1309	16353.129	10.0.0.1	224.0.0.9	RIPv2	146	Response
1310	16353.242	fe80::e95:11ff...	ff02::9	RIPng	186	Command Response, Version 1
1311	16355.998	fe80::e95:11ff...	ff02::1	ICMP...	110	Router Advertisement from 0c:95:11:9d:00:00
1312	16357.254	10.0.0.1	224.0.0.5	OSPF	78	Hello Packet
1313	16357.583	fe80::e95:11ff...	ff02::5	OSPF	90	Hello Packet

Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
Ethernet II, Src: Private 66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request/gratuitous ARP)

Захваченный трафик

Создадим проект в GNS3. Разместим и соединим устройства в соответствии с топологией. (рис. [-@fig:062]).



Проект

Присвоим адреса оконечным устройствам (рис. [-@fig:063]).

```

Hostname is too long. (Maximum 12 characters)

VPCS> ip 1000::a/64
PC1 : 1000::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 1000::a/64
DNS             :
ROUTER LINK-LAYER :
MAC             : 00:50:79:66:68:00
LPORT           : 20072
RHOST:PORT      : 127.0.0.1:20073
MTU             : 1500

VPCS> 
```

Присвоим адреса

Присвоим адреса оконечным устройствам (рис. [-@fig:064]).

```

Hostname is too long. (Maximum 12 characters)

VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 20074
RHOST:PORT           : 127.0.0.1:20075
MTU                 : 1500

VPCS> █

```

Присвоим адреса

На маршрутизаторах перейдем в режим конфигурирования и изменим имя устройства (рис. [-@fig:065]).

```

vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dmmosharov-gw-01
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dmmosharov-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] y█

```

Перейдем в режим конфигурирования

На маршрутизаторах перейдем в режим конфигурирования и изменим имя устройства (рис. [-@fig:066]).

```

vyos@vyos:~$ configure
set[edit]
vyos@vyos# set system host-name msk-dmmosharov-gw-02
[edit]
vyos@vyos# com
comment          commit          commit-confirm  compare
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dmmosharov-gw-02
[edit]
vyos@vyos# comm
comment          commit          commit-confirm
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] ☐

```

Перейдем в режим конфигурирования

На маршрутизаторах перейдем в режим конфигурирования и изменим имя устройства (рис. [-@fig:067]).

```

vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dmmosharov-gw-03
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dmmosharov-gw-03
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ ☐

```

Перейдем в режим конфигурирования

Настроим адреса на интерфейсах маршрутизаторов (рис. [-@fig:068]).

```

vyos@msk-dmmosharov-gw-01:~$ configure
[edit]
vyos@msk-dmmosharov-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-dmmosharov-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
vyos@msk-dmmosharov-gw-01# set service router-advert interface eth0 prefix 1000:
:/64
[edit]
vyos@msk-dmmosharov-gw-01# commit
[edit]
vyos@msk-dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-01# █

```

Настроим адреса

Настроим адреса на интерфейсах маршрутизаторов (рис. [-@fig:069]).

```

[edit]
vyos@msk-dmmosharov-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-dmmosharov-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8
[edit]
vyos@msk-dmmosharov-gw-02# set service router-advert interface eth0 prefix 1002:
:/64

Invalid value
Value validation failed
Set failed

[edit]
vyos@msk-dmmosharov-gw-02# commit
[edit]
vyos@msk-dmmosharov-gw-02# set service router-advert interface eth0 prefix 1002:
:/64
[edit]
vyos@msk-dmmosharov-gw-02# commit
[edit]
vyos@msk-dmmosharov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-02# █

```

Настроим адреса

Настроим адреса на интерфейсах маршрутизаторов (рис. [-@fig:070]).

```

vyos@msk-dmmosharov-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
vyos@msk-dmmosharov-gw-03# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@msk-dmmosharov-gw-03# delete interfaces ethernet eth
eth0 eth1 eth2 eth3 eth4 eth5 eth6 eth7 eth8 eth9
[edit]
vyos@msk-dmmosharov-gw-03# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-dmmosharov-gw-03# com
comment          commit          commit-confirm  compare
[edit]
vyos@msk-dmmosharov-gw-03# commit
[edit]
vyos@msk-dmmosharov-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-03# █

```

Настроим адреса

Убедимся, что появились адреса ближайших к ним маршрутизаторов (рис. [-@fig:071]).

```

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 1000::a/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT          : 20072
RHOST:PORT      : 127.0.0.1:20073
MTU            : 1500

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 1000::a/64
DNS            :
ROUTER LINK-LAYER : 0c:27:a7:8e:00:00
MAC            : 00:50:79:66:68:00
LPORT          : 20072
RHOST:PORT      : 127.0.0.1:20073
MTU            : 1500

VPCS> █

```

Убедимся, что появились адреса

Настроим маршрутизацию IPv4 (рис. [-@fig:073]).

```
vyos@msk-dmmosharov-gw-01:~$ ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.60 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.665 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.959 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.744 ms
^C
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 9ms
rtt min/avg/max/mdev = 0.665/1.493/3.604/1.223 ms
vyos@msk-dmmosharov-gw-01:~$ ping 20.0.0.1
connect: Network is unreachable
vyos@msk-dmmosharov-gw-01:~$ ping 20.0.0.2
connect: Network is unreachable
```

Настройка маршрутизации

Настроим маршрутизацию IPv4 (рис. [-@fig:074]).

```
you can check individual component licenses under /usr/share/c
vyos@msk-dmmosharov-gw-02:~$ configure
[edit]
vyos@msk-dmmosharov-gw-02# set protocols rip net
network          network-distance
[edit]
vyos@msk-dmmosharov-gw-02# set protocols rip network 20.0.0.0/
[edit]
vyos@msk-dmmosharov-gw-02# commit
[edit]
vyos@msk-dmmosharov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-02# █
```

Настройка маршрутизации

Настроим маршрутизацию IPv4 (рис. [-@fig:075]).

```
vyos@msk-dmmosharov-gw-03:~$ configure
[edit]
vyos@msk-dmmosharov-gw-03# set protocols rip net
network                network-distance
[edit]
vyos@msk-dmmosharov-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-dmmosharov-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-dmmosharov-gw-03# commit
[edit]
vyos@msk-dmmosharov-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-03#
```

Настройка маршрутизации

Проверим маршруты (рис. [-@fig:076]).

```
vyos@msk-dmmosharov-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=5.38 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.750 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.847 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.837 ms
^C
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 35ms
rtt min/avg/max/mdev = 0.750/1.952/5.376/1.977 ms
[edit]
vyos@msk-dmmosharov-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=2.14 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=0.930 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=0.658 ms
^C
--- 20.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 5ms
rtt min/avg/max/mdev = 0.658/1.243/2.142/0.645 ms
[edit]
vyos@msk-dmmosharov-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=3.06 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=1.18 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=1.20 ms
^C
--- 20.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 5ms
rtt min/avg/max/mdev = 1.176/1.813/3.061/0.883 ms
[edit]
vyos@msk-dmmosharov-gw-01#
```

Проверка

Создадим туннель IPv6 через сеть IPv4 (рис. [-@fig:077]).

```

vyos@msk-dmmosharov-gw-01:~$ configure
[edit]
vyos@msk-dmmosharov-gw-01# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-dmmosharov-gw-01# set interfaces tunnel tun0 sour
source-address      source-interface
[edit]
vyos@msk-dmmosharov-gw-01# set interfaces tunnel tun0 source-address 10.0.0.1
[edit]
vyos@msk-dmmosharov-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
[edit]
vyos@msk-dmmosharov-gw-01# set interfaces tunnel tun0 address 1001::1/64
[edit]
vyos@msk-dmmosharov-gw-01# commit
[edit]
vyos@msk-dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-01# █

```

Туннель

Создадим туннель IPv6 через сеть IPv4 (рис. [-@fig:078]).

```

vyos@msk-dmmosharov-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-dmmosharov-gw-02# set interfaces tunnel tun0 sour
source-address      source-interface
[edit]
vyos@msk-dmmosharov-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-dmmosharov-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-dmmosharov-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-dmmosharov-gw-02# commit
[edit]
vyos@msk-dmmosharov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-02# █

```

Туннель

Настроим статическую маршрутизацию IPv6 (рис. [-@fig:079]).

```

[edit]
vyos@msk-dmmosharov-gw-01# set protocols static route6 1002::0/64 next-hop 1001:
:2
[edit]
vyos@msk-dmmosharov-gw-01# commit
[edit]
vyos@msk-dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-01#

```

Настройка

Настроим статическую маршрутизацию IPv6 (рис. [-@fig:080]).

```

Done
[edit]
vyos@msk-dmmosharov-gw-02# set protocols static route6 1002::0/64 next-hop 1001:
:1
[edit]
vyos@msk-dmmosharov-gw-02# commit
[edit]
vyos@msk-dmmosharov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dmmosharov-gw-02#

```

Настройка

Проверим доступность оконечных устройств (рис. [-@fig:081]).

```

VPCS> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=2.180 ms
1002::a icmp6_seq=2 ttl=60 time=2.066 ms
1002::a icmp6_seq=3 ttl=60 time=2.345 ms
1002::a icmp6_seq=4 ttl=60 time=2.141 ms
^C
VPCS> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1  0.610 ms  0.425 ms  0.519 ms
 2 1001::2  1.548 ms  1.087 ms  1.304 ms
 3 1002::a  1.763 ms  1.579 ms  1.783 ms

VPCS>

```

Проверка доступности

Проверим доступность оконечных устройств (рис. [-@fig:082]).

```
VPCS> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=3.674 ms
1000::a icmp6_seq=2 ttl=60 time=2.240 ms
1000::a icmp6_seq=3 ttl=60 time=2.439 ms
1000::a icmp6_seq=4 ttl=60 time=2.310 ms
1000::a icmp6_seq=5 ttl=60 time=2.119 ms

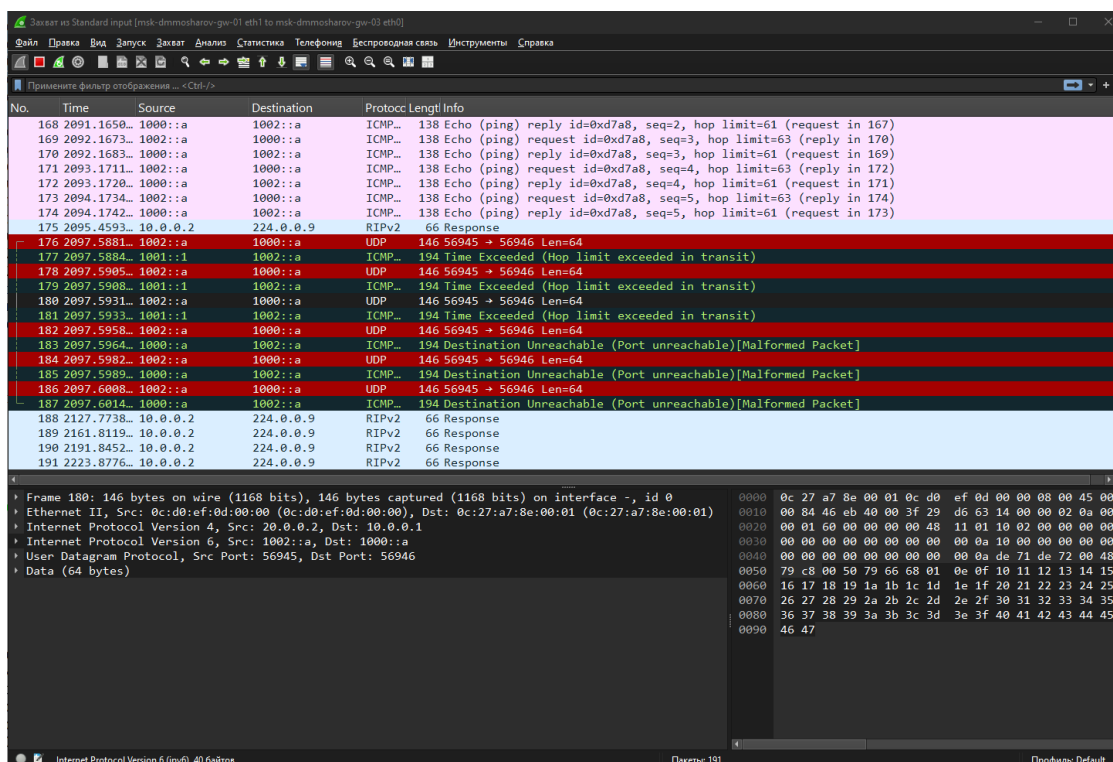
VPCS> trace 1000::a

trace to 1000::a, 64 hops max
 1 1002::1 0.656 ms 0.373 ms 0.280 ms
 2 1001::1 1.886 ms 1.216 ms 1.138 ms
 3 1000::a 1.601 ms 1.653 ms 1.611 ms

VPCS> █
```

Проверка доступности

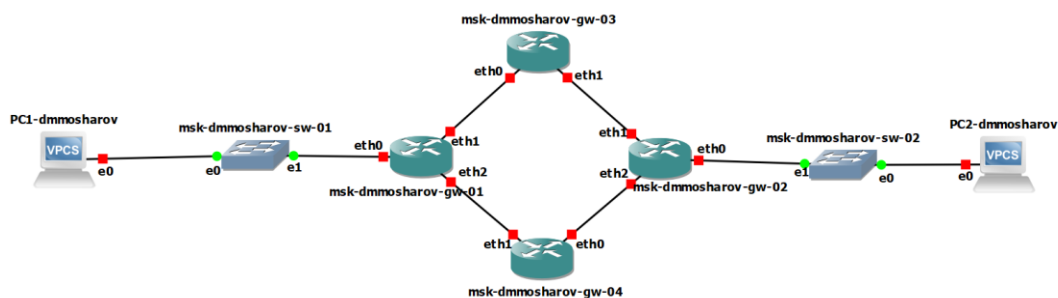
Проанализируем трафик (рис. [-@fig:083]).



Трафик

Самостоятельная работа

Построим топологию сети (рис. [-@fig:084]).



Топология сети

```

VPCS> ip 10.10.1.100/27 10.10.1.97/27
Checking for duplicate address...
PC1 : 10.10.1.100 255.255.255.224 gateway 10.10.1.97

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.10.1.100/27
GATEWAY    : 10.10.1.97
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10001
RHOST:PORT : 127.0.0.1:10002
MTU       : 1500

VPCS> █

```

Самостоятельная работа

```
VPCS> ip 10.10.1.70/28 10.10.1.65/28
Checking for duplicate address...
VPCS : 10.10.1.70 255.255.255.240 gateway 10.10.1.65

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.10.1.70/28
GATEWAY     : 10.10.1.65
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20063
RHOST:PORT  : 127.0.0.1:20064
MTU         : 1500

VPCS> 
```

Самостоятельная работа

```
frr# configure terminal
frr(config)# hostname msk-dmmosharov-gw01
msk-dmmosharov-gw01(config)# interface eth0
msk-dmmosharov-gw01(config-if)# ip address 10.10.1.97/27
msk-dmmosharov-gw01(config-if)# no shutdown
msk-dmmosharov-gw01(config-if)# exit
msk-dmmosharov-gw01(config)# interface eth1
msk-dmmosharov-gw01(config-if)# ip address 10.10.1.5/30
msk-dmmosharov-gw01(config-if)# no shutdown
msk-dmmosharov-gw01(config-if)# exit
msk-dmmosharov-gw01(config)# interface eth2
msk-dmmosharov-gw01(config-if)# ip address 10.10.1.9/30
msk-dmmosharov-gw01(config-if)# no shutdown
msk-dmmosharov-gw01(config-if)# exit
msk-dmmosharov-gw01(config)# exit
msk-dmmosharov-gw01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw01
service integrated-vtysh-config
!
interface eth0
 ip address 10.10.1.97/27
exit
!
interface eth1
 ip address 10.10.1.5/30
exit
!
interface eth2
 ip address 10.10.1.9/30
exit
!
end
msk-dmmosharov-gw01#
```

Самостоятельная работа

```
frr# configure terminal
frr(config)# hostname msk-dmmosharov-gw-02
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# ip address 10.10.1.65/28
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# interface eth1
msk-dmmosharov-gw-02(config-if)# ip address 10.10.1.17/30
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# interface eth2
msk-dmmosharov-gw-02(config-if)# ip address 10.10.1.33/30
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.10.1.65/28
exit
!
interface eth1
 ip address 10.10.1.17/30
exit
!
interface eth2
 ip address 10.10.1.33/30
exit
!
end
msk-dmmosharov-gw-02#
```

Самостоятельная работа

```
frr# configure terminal
frr(config)# hostname msk-dmmosharov-user-gw-03
msk-dmmosharov-user-gw-03(config)# interface eth0
msk-dmmosharov-user-gw-03(config-if)# ip address 10.10.1.5/30
msk-dmmosharov-user-gw-03(config-if)# no shutdown
msk-dmmosharov-user-gw-03(config-if)# exit
msk-dmmosharov-user-gw-03(config)# interface eth1
msk-dmmosharov-user-gw-03(config-if)# ip address 10.10.1.17/30
msk-dmmosharov-user-gw-03(config-if)# no shutdown
msk-dmmosharov-user-gw-03(config-if)# exit
msk-dmmosharov-user-gw-03(config)# exit
msk-dmmosharov-user-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-user-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-user-gw-03
service integrated-vtysh-config
!
interface eth0
    ip address 10.10.1.5/30
exit
!
interface eth1
    ip address 10.10.1.17/30
exit
!
end
msk-dmmosharov-user-gw-03#
```

Самостоятельная работа

```
frr# configure terminal
frr(config)# hostname msk-dmmosharov-gw-04
msk-dmmosharov-gw-04(config)# interface eth0
msk-dmmosharov-gw-04(config-if)# ip address 10.10.1.100/27
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# interface eth1
msk-dmmosharov-gw-04(config-if)# ip address 10.10.1.9/30
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.10.1.100/27
exit
!
interface eth1
 ip address 10.10.1.9/30
exit
!
end
msk-dmmosharov-gw-04#
```

Самостоятельная работа

```
VPCS> ip 2001:db8:1:1::a/64
PC1 : 2001:db8:1:1::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 2001:db8:1:1::a/64
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT         : 10001
RHOST:PORT     : 127.0.0.1:10002
MTU           : 1500

VPCS> █
```

Самостоятельная работа

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.10.1.70/28
GATEWAY    : 10.10.1.65
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20063
RHOST:PORT : 127.0.0.1:20064
MTU        : 1500

VPCS> ip 2001:db8:1:6::a/64
PC1 : 2001:db8:1:6::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 2001:db8:1:6::a/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT          : 20063
RHOST:PORT     : 127.0.0.1:20064
MTU            : 1500

VPCS> █
```

Самостоятельная работа

```
msk-dmmosharov-gw01# configure terminal
msk-dmmosharov-gw01(config)# ipv6 forwarding
msk-dmmosharov-gw01(config)# interface eth0
msk-dmmosharov-gw01(config-if)# ipv6 address 2001:db8:1:1::1/64
msk-dmmosharov-gw01(config-if)# no ipv6 nd suppress-ra
msk-dmmosharov-gw01(config-if)# ipv6 nd prefix 2001:db8:1:1::/64
msk-dmmosharov-gw01(config-if)# no shutdown
msk-dmmosharov-gw01(config-if)# exit
msk-dmmosharov-gw01(config)# interface eth2
msk-dmmosharov-gw01(config-if)# ipv6 address 2001:db8:1:2::1/64
msk-dmmosharov-gw01(config-if)# no shutdown
msk-dmmosharov-gw01(config-if)# exit
msk-dmmosharov-gw01(config)# exit
msk-dmmosharov-gw01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw01
service integrated-vtysh-config
!
interface eth0
 ip address 10.10.1.97/27
 ipv6 address 2001:db8:1:1::1/64
 ipv6 nd prefix 2001:db8:1:1::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.10.1.5/30
exit
!
interface eth2
 ip address 10.10.1.9/30
 ipv6 address 2001:db8:1:2::1/64
exit
!
end
msk-dmmosharov-gw01#
```

Самостоятельная работа

```

msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# ipv6 forwarding
msk-dmmosharov-gw-02(config)# interface eth0
msk-dmmosharov-gw-02(config-if)# ipv6 address 2001:db8:1::1/64
msk-dmmosharov-gw-02(config-if)# no ipv6 nd suppress-ra
msk-dmmosharov-gw-02(config-if)# ipv6 nd prefix 2001:db8:1::
    X:X::X:X/M IPv6 prefix
msk-dmmosharov-gw-02(config-if)# ipv6 nd prefix 2001:db8:1::/64
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# interface eth1
msk-dmmosharov-gw-02(config-if)# ipv6 address 2001:db8:1:4::1/64
msk-dmmosharov-gw-02(config-if)# no shutdown
msk-dmmosharov-gw-02(config-if)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-02
service integrated-vtysh-config
!
interface eth0
    ip address 10.10.1.65/28
    ipv6 address 2001:db8:1::1/64
    ipv6 nd prefix 2001:db8:1::/64
    no ipv6 nd suppress-ra
exit
!
interface eth1
    ip address 10.10.1.17/30
    ipv6 address 2001:db8:1:4::1/64
exit
!
interface eth2
    ip address 10.10.1.33/30
exit
!
end
msk-dmmosharov-gw-02# █

```

Самостоятельная работа

```
msk-dmmosharov-user-gw-03# configure terminal
msk-dmmosharov-user-gw-03(config)# ipv6 forwarding
msk-dmmosharov-user-gw-03(config)# interface eth0
msk-dmmosharov-user-gw-03(config-if)# ipv6 address 2001:db8:1:2::1/64
msk-dmmosharov-user-gw-03(config-if)# no shutdown
msk-dmmosharov-user-gw-03(config-if)# exit
msk-dmmosharov-user-gw-03(config)# interface eth1
msk-dmmosharov-user-gw-03(config-if)# ipv6 address 2001:db8:1:3::1/64
msk-dmmosharov-user-gw-03(config-if)# no shutdown
msk-dmmosharov-user-gw-03(config-if)# exit
msk-dmmosharov-user-gw-03(config)# exit
msk-dmmosharov-user-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-user-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-user-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.10.1.5/30
 ipv6 address 2001:db8:1:2::1/64
exit
!
interface eth1
 ip address 10.10.1.17/30
 ipv6 address 2001:db8:1:3::1/64
exit
!
end
msk-dmmosharov-user-gw-03#
```

Самостоятельная работа

```
msk-dmmosharov-gw-04# configure terminal
msk-dmmosharov-gw-04(config)# interface eth0
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# ipv6 forwarding
msk-dmmosharov-gw-04(config)# interface eth0
msk-dmmosharov-gw-04(config-if)# ipv6 address 2001:db8:1:5::1/64
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# interface eth1
msk-dmmosharov-gw-04(config-if)# ipv6 address 2001:db8:1:3::1/64
msk-dmmosharov-gw-04(config-if)# no shutdown
msk-dmmosharov-gw-04(config-if)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dmmosharov-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.10.1.100/27
 ip address 10.10.1.33/30
 ipv6 address 2001:db8:1:5::1/64
exit
!
interface eth1
 ip address 10.10.1.9/30
 ipv6 address 2001:db8:1:3::1/64
exit
!
end
msk-dmmosharov-gw-04#
```

Самостоятельная работа

```
msk-dmmosharov-gw01# configure terminal
msk-dmmosharov-gw01(config)# router rip
msk-dmmosharov-gw01(config-router)# version 2
msk-dmmosharov-gw01(config-router)# network eth0
msk-dmmosharov-gw01(config-router)# network eth1
msk-dmmosharov-gw01(config-router)# network eth2
msk-dmmosharov-gw01(config-router)# exit
msk-dmmosharov-gw01(config)# exit
msk-dmmosharov-gw01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw01#
```

Самостоятельная работа

```
msk-dmmosharov-gw-02# configure terminal
msk-dmmosharov-gw-02(config)# router rip
msk-dmmosharov-gw-02(config-router)# version 2
msk-dmmosharov-gw-02(config-router)# network eth0
msk-dmmosharov-gw-02(config-router)# network eth1
msk-dmmosharov-gw-02(config-router)# network eth2
msk-dmmosharov-gw-02(config-router)# exit
msk-dmmosharov-gw-02(config)# exit
msk-dmmosharov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-02#
```

Самостоятельная работа

```
msk-dmmosharov-gw-04# configure terminal
msk-dmmosharov-gw-04(config)# router rip
msk-dmmosharov-gw-04(config-router)# version 2
msk-dmmosharov-gw-04(config-router)# network eth0
msk-dmmosharov-gw-04(config-router)# network eth1
msk-dmmosharov-gw-04(config-router)# exit
msk-dmmosharov-gw-04(config)# exit
msk-dmmosharov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dmmosharov-gw-04#
```

Самостоятельная работа

```

msk-dmmosharov-gw01# ip route rip
% Unknown command: ip route rip
msk-dmmosharov-gw01# show ip route rip
msk-dmmosharov-gw01# ip rip
% Unknown command: ip rip
msk-dmmosharov-gw01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.10.1.4/30      0.0.0.0          1 self              0
C(i) 10.10.1.8/30      0.0.0.0          1 self              0
C(i) 10.10.1.96/27     0.0.0.0          1 self              0
msk-dmmosharov-gw01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 23 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2    2
    eth1           2    2
    eth2           2    2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
  Distance: (default is 120)
msk-dmmosharov-gw01# 

```

Самостоятельная работа

```

msk-dmmosharov-gw-02# show ip route rip
msk-dmmosharov-gw-02# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.10.1.16/30     0.0.0.0           1 self              0
C(i) 10.10.1.32/30     0.0.0.0           1 self              0
C(i) 10.10.1.64/28     0.0.0.0           1 self              0
msk-dmmosharov-gw-02# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 16 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2    2
    eth1           2    2
    eth2           2    2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
  Distance: (default is 120)
msk-dmmosharov-gw-02#

```

Самостоятельная работа

```

msk-dmmosharov-user-gw-03# show ip route rip
msk-dmmosharov-user-gw-03# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
    (n) - normal, (s) - static, (d) - default, (r) - redistribute,
    (i) - interface

    Network          Next Hop          Metric From          Tag Time
C(i) 10.10.1.4/30    0.0.0.0           1 self              0
C(i) 10.10.1.16/30  0.0.0.0           1 self              0
msk-dmmosharov-user-gw-03# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 29 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2    2
    eth1           2    2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
  Distance: (default is 120)
msk-dmmosharov-user-gw-03#

```

Самостоятельная работа

```

msk-dmmosharov-gw-04# show ip route rip
msk-dmmosharov-gw-04# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
    (n) - normal, (s) - static, (d) - default, (r) - redistribute,
    (i) - interface

    Network          Next Hop          Metric From          Tag Time
C(i) 10.10.1.8/30    0.0.0.0           1 self              0
C(i) 10.10.1.32/30   0.0.0.0           1 self              0
C(i) 10.10.1.96/27   0.0.0.0           1 self              0
msk-dmmosharov-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 24 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2     2
    eth1           2     2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
  Distance: (default is 120)
msk-dmmosharov-gw-04# █

```

Самостоятельная работа

Выводы

В результате выполнения лабораторной работы были изучены принципы маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.